

Referee Report

Paper: *Magnitude, Not Shape: Volatility Beats Downside- and Tail-Risk Metrics at Forecasting Drawdown* (anonymous submission, 7 pp.)

Target venue: ACM ICAIF '26 main track — Milan, Nov 14–17, 2026. Submission deadline **Aug 2, 2026 (AoE)**; **8-page cap including figures and references**; double-blind; **no rebuttal period**; **no supplementary materials or appendices accepted**.

Review stance: senior program-committee / area-chair-grade review, written to be actionable before the deadline.

1. Summary of the submission

The paper asks whether any of nine downside- or tail-risk metrics (downside deviation, VaR, ES, negative realized semibeta, downside beta, lower-tail dependence, a GDA proxy, the Viole–Nawrocki UPM/LPM ratio, and a per-asset Hill tail exponent) beats plain trailing volatility at forecasting 90-day forward maximum drawdown in the cross-section. The evaluation runs on five test beds — three survivorship-free crypto regimes (2016 calm, 2017 bull, 2018 crash) and two survivorship-biased S&P 500 windows (1994–99, 2005–09) — scored by per-date cross-sectional Spearman correlations with paired block-bootstrap CIs and Benjamini–Hochberg correction, with a pre-fixed generalizability bar (must beat volatility on at least one crypto and one equity bed).

Headline results: no metric clears the bar (§3). The anatomy (§4) is that only tail-*magnitude* measures add a small partial-correlation increment beyond volatility (VaR up to +0.19), while shape measures add ≈ 0 or negative — yet shape metrics do win on genuinely shape-sensitive targets (Table 4), so they are faithful, just aimed at the wrong target. Volatility wins because it is persistent, though a disjoint-window mediation test shows forward volatility is not a sufficient statistic (§5.1). A pre-registered composite $z(\text{vol})+z(\text{VaR})$ adds nothing significant out-of-sample except on crypto-bull (§5.2). An LLM discovery agent and a 20,000-configuration sweep inflate validation scores while the locked test plateaus at the volatility baseline — an agentic data-snooping caution (§5.3). Ablations (§6) show each fair-field discipline changes what one would report but not the verdict.

2. Overall assessment

Recommendation as submitted: Weak Accept (6/10), confidence 4/5. After the mandatory fixes below: Accept-grade (8/10).

This is an unusually disciplined negative-result paper with a clean thesis, real statistical hygiene, and a timely agentic-search angle that fits ICAIF well. The writing is confident and clear, and the internal numerical consistency across tables is excellent (I checked; see S3).

However, the paper currently contains **two correctness-level internal inconsistencies** (M1, M2): its own Table 2 and Figure 1 contradict the headline claim that "no systematic co-movement metric adds signal," and the metric taxonomy is inconsistent about the Viole–Nawrocki ratio. At a venue with **no rebuttal**, a reviewer who catches either one has no way to hear your explanation — these are the difference between acceptance and rejection, and both are fixable in a day or two. Fixed properly, they *strengthen* the thesis rather than weaken it (see M1's suggested reframing).

Dimension	As submitted	Achievable by Aug 2
Fit to ICAIF main track	5/5	5/5

Technical soundness	3/5	4.5/5
Novelty / contribution	4/5	4/5
Presentation & writing	4/5	4.5/5
Reproducibility	3.5/5	4.5/5

3. Strengths

S1 — Adversarial design done honestly. Pre-fixed generalizability bar (§3), BH correction, paired block-bootstrap CIs, an MDE/power analysis on the thin beds, a placebo permutation test, and §6 ablations that quantify what each discipline (survivorship-free data, multiple-testing correction, locked holdout) contributes. Very few empirical-finance submissions do half of this.

S2 — The anatomy is a real contribution, and Table 4 is the paper's best move. Re-scoring every metric against shape-sensitive targets (forward skew, downside fraction, tail-exceedance frequency) and showing the picture inverts preempts the obvious "your daily adaptations are strawmen" objection. "Shape metrics forecast shape; shape is not what drives drawdown" is a memorable, defensible empirical law.

S3 — Internal numerical coherence. Table 5's (trail→dd) column exactly reproduces Table 1's volatility row (0.62/0.36/0.38/0.61/0.56); Table 6's vol column matches Table 1; Table 8's head-to-head entries match Table 1 differences (e.g., semibeta crypto-18: $0.31 - 0.38 \approx -0.067$). The pipeline is evidently one code path. This is what reviewers' spot-checks look for, and it passes.

S4 — The agentic data-snooping experiment is timely and well-framed for ICAIF. "An autonomous agent removes the last brake on trial count, namely analyst effort, so a locked holdout scored exactly once becomes mandatory, not merely advisable" is quotable and policy-relevant. The observation that the non-LLM sweep reproduces the plateau ("the danger is the cheap search, not the language model") is a sharp, honest point.

S5 — Rare candor. The limitations section is real, and §7's note that two earlier over-claims were stress-tested and withdrawn signals exactly the epistemic culture the paper advocates.

S6 — Reproducibility intent. Anonymized repository with code, data pointers, and result logs (footnote 1) — assuming the pre-registration artifact lands there too (see M3).

S7 — Writing. Clear, compact, confident. "The verdict is blunt." "Volatility is the workhorse." The prose carries a complex design without jargon debt.

4. Major issues (must fix before submission)

M1 — Your own Table 2 contradicts the headline claim "no systematic co-movement metric adds signal." (Correctness)

Table 2 reports negative semibeta with **BH-surviving positive increments on three beds: +0.08* (crypto-17), +0.13* (crypto-18), and +0.05* (equity-05)**, where * is defined as "survives BH ($q = 0.05$, positive)." Figure 1's own whiskers corroborate this — the semibeta CIs sit clear of zero in those panels. Yet:

- §4 asserts "no systematic co-movement metric adds signal (Figure 1 groups both axes)";
- Figure 1's in-plot title says "only own-asset magnitude adds signal" and its caption repeats "no systematic co-movement metric (red) adds signal";

- the abstract says "Every tail-shape metric adds nothing or subtracts" — and Table 2's caption classifies semibeta as tail shape (rows 4–9).

This is worse than a typo: by the paper's own generalizability bar (BH-significant on ≥ 1 crypto and ≥ 1 equity bed), semibeta's partial increment qualifies as generalizable incremental signal. The §4 text even walks through the exact logic for VaR ("worse standalone, but the partial correlation isolates its small orthogonal component") that a hostile reviewer will apply to semibeta.

Fix (and it strengthens the paper). First verify against your result logs that the three stars are not a table-generation error. If they are real — which Figure 1 suggests — own them and sharpen the taxonomy: realized negative semibeta is built from down-down semicovariances, so it *carries downside magnitude*; it is not a magnitude-free shape measure. Restate the law as: **the only BH-surviving increments come from metrics that embed downside magnitude (VaR, ES, downside deviation, and — weakly — negative semibeta), while every magnitude-free shape measure (UPM/LPM, Hill, lower-tail dependence, GDA, downside beta) adds ≈ 0 or negative.** That is still "magnitude, not shape," arguably in cleaner form, and it converts an inconsistency into a supporting observation. Update the abstract, §4, and Figure 1's title/caption together.

M2 — The taxonomy is internally inconsistent about the Viole–Nawrocki ratio, and Figure 1 encodes the inconsistency. (Correctness)

The VN UPM/LPM ratio is computed from an asset's own return distribution — it is an own-asset shape measure. §2's list is consistent with that (it names VN and Hill, then "four systematic co-movement measures"). But §4 calls the Hill exponent "the one own-asset shape metric," and Figure 1 colors VN UPM/LPM **red** (systematic/co-movement), leaving Hill as the lone orange row. Two of the paper's three claimed metric families disagree with its own §2 definitions.

Fix. Either (a) recolor VN orange in Figure 1 and rewrite the §4 sentence as "the two own-asset shape metrics (the partial-moment ratio and the Hill exponent) are themselves ≈ 0 /negative," or (b) if your implementation actually computes the VN ratio against the market, say so explicitly in §2 — but then the intro's framing of UPM/LPM as an own-asset asymmetry measure needs adjusting. Option (a) is almost certainly what you mean, and the "magnitude beats shape within the own-asset family" conclusion survives unchanged with two data points instead of one.

M3 — "Pre-registered" and "a priori" need an artifact and precise staging; §5.2 currently undercuts itself. (Credibility)

§5.2: "VaR was chosen **a priori** as the magnitude metric **surviving correction on all five beds.**" Choosing a component *because it survived your main tests* is by definition post hoc relative to those tests. What you evidently mean — and what the design supports — is that the composite was frozen after the exploratory beds but **before the locked panel was opened**, which is why "the fully out-of-sample evidence is the locked panel." Say exactly that and delete "a priori."

Separately, the paper claims pre-registration three times ("We pre-registered the main tests...", "We pre-registered it") with no checkable artifact. Reviewers discount unverifiable pre-registration claims to zero. Add to footnote 1: a dated pre-registration document (or timestamped commit hash) in the anonymized repository specifying the frozen tests, the generalizability bar, and the composite.

M4 — The placebo fails to reach ≈ 0 on one bed, and the paper does not say which or why. (Soundness / sanity check)

§2: "A placebo that permutes forward drawdown within each date drives the volatility correlation to ≈ 0 on **four of five** beds." A within-date permutation destroys the cross-sectional pairing by construction, so the mean correlation should be ≈ 0 on every bed, up to thin-sample noise. As written, this reads like an

unexplained failed sanity check — the single most invitation-to-reject sentence in the paper, because a reviewer's imagination will fill the gap with "pipeline bug."

Fix. One sentence: name the bed, give the placebo value, and attribute it (presumably a 7-date crypto bed where the permutation-null distribution is wide — if so, report the permutation-null CI and show the observed placebo sits inside it). If you cannot explain it, that is a bug hunt you must do before submission.

M5 — §5.3's numbers do not support "often straddling zero," and "never exceeds noise" is asserted without a test. (Statistics / interpretation)

The random-repartition family's locked-test gap is "+0.035 (range **+0.014 to +0.072**)" — strictly positive in every run. Only the out-of-time family straddles zero (−0.028 to +0.046). "Always small and often straddling zero" mischaracterizes half your own robustness grid, and Figure 5's caption claim that the edge "never exceeds noise" has no accompanying test.

Fix, two parts. (a) Back "never exceeds noise" with a number — e.g., a t-statistic across the 12 panels × 3 seeds (clustered by panel) for each family. (b) Interpret the uniformly positive random-split gap as what it almost certainly is: **temporal leakage** — order-ignoring repartitions interleave validation and test dates, so persistence leaks across the split. That is not a nuisance to be glossed; it is a *second data-snooping lesson* that strengthens the section: naive random CV manufactures a phantom edge in persistent panels, and only chronological holdouts are honest. One rewritten sentence plus one added sentence turns your weakest paragraph into one of your strongest.

M6 — The LLM discovery loop — the paper's most ICAIF-relevant element — is under-specified to the point of irreproducibility. (Completeness)

§5.3 tells us an "LLM agent rewriting a scoring function" ran against a 20,000-configuration sweep, and a "general nonlinear candidate generator (1,500 candidates per run)" ran across 12 panels × 3 seeds. Missing: which model and version; the agent protocol (what it sees, what it may rewrite, iteration count, stopping rule, temperature); how the 20,000-config space is defined; what function family the 1,500 candidates span; and cost. The abstract and keywords foreground "LLM agents"; reviewers will go looking for the machinery and find one paragraph.

Fix. ICAIF '26 accepts **no appendices**, so this must be ~0.3–0.5 page in the main text (compact protocol paragraph or algorithm box) with full detail in the anonymized repo. You have the space: the paper is 7 pages against an 8-page cap that includes references.

M7 — Inference on the crypto beds is thinner than some of the language admits. (Statistics)

With 7–9 formation dates and length-3 blocks, the block bootstrap rests on ~3 effective blocks; CI coverage at that size is fragile. The MDE analysis and the "suggestive" framing are the right instinct — but apply them uniformly: the abstract's flagship claim ("survives BH correction on all five beds") leans on those beds with no qualifier, and the composite's only win (Table 6: +0.05 [0.02, 0.07] on crypto-17) is on a 7–9-date bed yet is not run through the same MDE caveat that §2 applies to VaR. Also: justify block length 3, and add one sentence showing robustness to block length (1/2/4 or a stationary bootstrap).

M8 — Survivorship framing: accurate in §2, oversold in the abstract; one overgeneralization. (Framing)

The abstract's "On a survivorship-free ... evaluation" describes two of five beds inaccurately — the equity beds are survivorship-biased, as §2 admits at length. Reviewers read abstracts adversarially; a phrase like "survivorship-free where survivorship can be fixed (crypto), with equity beds as a cross-asset check" costs six words and removes the attack surface. Likewise "no free-data equity bed is genuinely survivorship-free"

generalizes from checking a single dataset (Kaggle); soften to what you verified. Finally, consider one sentence applying §6's direction-of-bias finding to equity: survivors-only data flattered shape and inflated the magnitude edge on crypto-2018, so the small equity increments are plausibly upper bounds and shape's equity penalty a lower bound — the conservative direction for your thesis. Saying so preempts the "your equity beds are biased" review with "yes, and the bias runs against us."

M9 — Related work omits the literature the keywords promise: LLM/agentic discovery in finance. (Positioning)

"Our novelty is to show an autonomous agent reaches the snooping frontier at near-zero cost" is uncited against the 2023–26 wave of LLM-driven alpha mining, automated factor/strategy discovery, and agentic quant-research systems — exactly the papers ICAIF reviewers know. Three to five citations plus one contrast sentence ("prior agentic-discovery work reports validation-selected wins; we show the locked-test fate of those wins") both protects the novelty claim and sharpens it. Consider also Sullivan–Timmermann–White (1999) alongside White (2000) in the snooping canon.

M10 — "Powered null" would be stronger as an equivalence statement. (Statistics, cheap upgrade)

"The observed shape increments sit at or below [the 80%-power MDE of 0.034–0.042], so 'shape adds nothing' is a powered null" is an informal power argument. The cleaner statement is an equivalence bound: e.g., "on the equity beds, the 95% CIs on every shape increment exclude effects larger than ≈ 0.04 " (or a TOST). One sentence converts "we would have seen it" into "we can bound it," and preempts the reflexive absence-of-evidence objection.

5. Minor issues

- Table 6 caption is wrong as stated.** "Significant only on crypto-bull" — but equity-94's difference is -0.008 [-0.011 , -0.004], whose CI excludes zero: significantly *negative*. Say "significantly positive only on crypto-bull; significantly negative (trivially small) on equity-94." The negative cell helps your thesis — report it. Also define the locked panel in the caption (2017→2018-crash, per Figure 5).
- Mediation design under-specified (§5.1, Figure 2).** Define the "disjoint, predictive window" precisely: which 30 days measure the mediator (forward volatility) and which window measures the drawdown. It is the paper's subtlest test and currently cannot be reconstructed from the text.
- Hill "n/e — not estimable per date on equity":** give the reason in one clause (tail-observation count? fit-quality filter?). Two tables carry n/e cells with no explanation.
- "Within each metric family (head-to-head and partials, ~45 cells each)"** — "family" is overloaded: Figure 1 uses "families" for the three metric groups. Rename the BH grouping ("test battery") and add a clause on why BH runs within battery rather than globally.
- "Validation-selected winners beat volatility on the locked test only 43% of the time"** — 43% of *what*? Search intensities, candidates, runs? Name the denominator.
- "A positive slope, suggestive when clustered by bed"** — report the clustered *t* or CI. "Suggestive" without a number invites the question.
- Equal-weight market benchmark** for the four co-movement metrics is nonstandard (value-weight is the convention). One sentence of justification or a robustness clause.
- Cost model detail (§5.2):** 20 bps per what (one-way? round trip?), rebalance frequency, and tercile weighting. Two clauses.

9. **Bed statistics are missing.** Assets per date and formation dates per bed appear nowhere ("144 dates" pooled appears once, in §4). A one-line summary per bed is needed to interpret power (relates to M7).
10. **Trailing-window choice (180 d) is unexamined** — the horizon sweep varies only the forward side. One robustness clause (e.g., 120 d / 250 d) or list it in Limitations.
11. **Kurtosis sentence invites misreading (§4):** "crypto increments 0.12–0.19, excess kurtosis 8–21, vs equity 0.03–0.08" — the equity range is increments, not kurtosis, but the parallel structure suggests otherwise. Give equity's kurtosis or restructure.
12. **Figure 1 label "GDA vol-down"** vs the text's "disappointment-aversion proxy" — align naming; define "VN" in the caption.
13. **Bed abbreviations (cr-16, eq 94)** are used in Table 1's header before being defined anywhere. Add to the caption.
14. **Abstract is ~240 words** and front-loads the three disciplines before the question. 170–190 words would sharpen it; the current second sentence ("The premise is always the same.") can absorb the next two.
15. **"Not economically free" (abstract, §1)** is ambiguous — it scans as "costly." You mean "does not translate into economic gains net of costs"; say that.
16. **"Delisting-reputed price dump"** — awkward coinage; "a public price dump reputed to include delisted stocks."
17. **§7 "retracted it here"** — "corrected in this version" reads better double-blind; "retracted" implies a published prior and hints at submission history.
18. **In-figure titles editorialize** ("Automated search inflates validation, not the locked test"; "only own-asset magnitude adds signal"). ACM tolerates this, but neutral in-figure titles with argumentative captions avoid the "advocacy graphics" reflex in skeptical reviewers — and the Figure 1 title must change anyway per M1.
19. **Page budget:** content ends early on p.6; references end mid-p.7. Against an 8-page cap including references you have ≈1.3 pages. Spend them on M6, M2, and minor 9 — do not leave them empty.

6. Line-by-line proofreading

The manuscript is typographically clean — I found exactly one true spelling error. Full list of mechanical edits:

- p.5, §5.4: "for the stricter $\leq -90\%$ **cutof**" → "**cutoff**". (The only outright typo.)
- p.2, §2: "which assets to **derisk**" → "de-risk" (or "trim").
- p.1, §1: "Each rests on the same premise, that asymmetry or tail structure..." → "...the same premise: that asymmetry..."
- p.1, abstract and §1 both start sentences with "And" ("And multiple-testing correction keeps..."); once is a choice, twice is a tic.
- p.3, §4: "Within crypto, splitting dates at their cross-sectional-median volatility, the increment rises..." — dangling modifier; "When crypto dates are split at the cross-sectional median of volatility, the increment rises..."
- p.3, Figure 1 caption: "grouped on both **axes** a skeptic cares about" — the color encoding is not an axis; "along the two dimensions a skeptic cares about."
- p.6–7, ref [12]: Engle (1982)'s title is "...variance of **United Kingdom** inflation" (the entry prints "U.K."); match the original.
- p.7, ref [16]: the Harvey-Liu-Zhu title really does begin with an ellipsis ("... and the Cross-Section of Expected Returns") — correct as printed; protect it from copy-editing "fixes" with a brace in the .bib.

- Table 8 mixes CI-bearing and bare entries; the caption's "where estimated" covers it, but either show all three CIs or mark significance consistently with the other tables' */† convention.
- Consistency spot-checks that pass: "crypto-bull" hyphenation, "z(vol) + z(VaR)" notation, §-references, "80%-power MDE," Table 5 vs Table 1 values, Table 6 vs Table 1 values.

7. Questions a reviewer will ask — preempt them in the text

1. Are the three semibeta stars in Table 2 correct? If so, how does "no systematic co-movement metric adds signal" survive? (M1)
2. Is the VN ratio computed on own-asset returns or against the market? (M2)
3. Which bed fails the placebo, and why? (M4)
4. Which LLM, what protocol, how many agent iterations vs. sweep configurations, and what does N count in Table 7? (M6, minor 5)
5. How many assets per date and formation dates per bed? Are the 90-day forward windows overlapping, and how sensitive are the CIs to block length? (M7, minor 9)
6. Where is the pre-registration artifact? (M3)
7. Does the composite's crypto-bull win clear the same MDE bar you apply to VaR's crypto claims? (M7)
8. Why equal-weight rather than value-weight market for the co-movement metrics? (minor 7)
9. Why is the Hill exponent not estimable per date on equity? (minor 3)
10. The random-split family's test gap is uniformly positive — leakage? (M5)

8. Venue compliance check (ICAIF '26 official CFP)

- **Deadline: Aug 2, 2026, AoE — five days from today.** Notification Sep 27; conference Nov 14–17, Bocconi University, Milan.
- **8 pages maximum including figures and references; over-length papers rejected without review.** Current draft: 7 pages → ≈1.3 pages of headroom for the M6/M2/minor-9 additions.
- **No supplementary materials or appendices accepted.** All protocol detail must live in the main text or the anonymized repo (repo link in footnote is the standard workaround and already present).
- **Double-blind:** anonymous byline, "Anon." running heads, anonymized repo, clean PDF metadata (no author fields) — compliant. Keep self-references third-person if any are added.
- **Template version:** the PDF was built with acmart v1.83 (2022). The CFP asks for the *latest* ACM template; acmart is now well past 1.90. Recompiling with the current version is a five-minute insurance policy against a desk-check nit.
- **No rebuttal period.** Every objection above must be preempted in the PDF; nothing can be explained later. This is why M1–M4 are existential rather than cosmetic.
- Administrative: max six submissions per author; the author list is frozen at submission.

9. Prioritized revision plan (five days)

Day 1 — correctness (blocking).

- ☐ Re-run/verify Table 2's semibeta stars against result logs; reconcile abstract + §4 + Figure 1 title/caption via the magnitude-bearing reframing (M1).
- ☐ Fix VN classification in Figure 1 and the "one own-asset shape metric" sentence (M2).
- ☐ Identify the placebo bed; add the one-sentence explanation or fix the bug (M4).

Day 2 — credibility.

- ☐ Rewrite the §5.2 "a priori" sentence; add the pre-registration artifact/commit hash to footnote 1 (M3).
- ☐ Rewrite the "straddling zero" sentence; add the clustered test behind "never exceeds noise"; add the leakage interpretation (M5).
- ☐ Fix Table 6's caption (minor 1).

Day 3 — content.

- ☐ Add the LLM/sweep protocol paragraph (M6).
- ☐ Add bed statistics (minor 9) and the block-length robustness clause (M7).
- ☐ Add 3–5 LLM-agents-in-finance citations plus a contrast sentence (M9).

Day 4 — polish.

- ☐ Tighten the abstract; fix the survivorship phrasing (M8); add the equivalence sentence (M10).
- ☐ Sweep minors 2–18 and the line edits of §6 (including "cutoff").

Day 5 — freeze.

- ☐ Full re-read; re-verify every cross-reference and table/text number; confirm page count ≤ 8 ; submit early (AoE deadlines invite server-rush accidents).

10. Suggested drop-in rewrites

- **Figure 1 in-plot title:** "only own-asset magnitude adds signal" → "only magnitude-bearing metrics add signal beyond volatility."
- **§4:** "while no systematic co-movement metric adds signal" → "among systematic metrics, only negative semibeta — which embeds downside magnitude through its semicovariance construction — shows a small BH-surviving increment (+0.05 to +0.13 on three beds); the magnitude-free co-movement measures add nothing."
- **Abstract:** "Every tail-shape metric adds nothing or subtracts." → "Every magnitude-free shape metric adds nothing or subtracts."
- **§5.2:** "VaR was chosen a priori as the magnitude metric surviving correction on all five beds, and the weights are untuned, so the fully out-of-sample evidence is the locked panel." → "VaR was fixed as the composite's second term before the locked panel was opened — it was the only metric whose increment survived correction on all five exploratory beds — and the weights are untuned, so the locked panel is fully out-of-sample."
- **§5.3:** "always small and often straddling zero" → "small throughout: the out-of-time gaps straddle zero (−0.028 to +0.046), while the random-repartition gaps are uniformly positive (+0.014 to +0.072) — the signature of temporal leakage when validation dates interleave with test dates, and itself a warning against order-ignoring cross-validation in persistent panels."
- **Abstract:** "The tiny magnitude edge is not economically free either." → "The tiny magnitude edge also fails to pay: a cost-aware backtest and a pre-registered composite add nothing over volatility outside fat-tailed crypto."

11. Verdict

This is a paper I would fight for in a PC discussion — *after* Day 1 and Day 2 of the plan above. The question is well-posed, the evaluation is more honest than most of the literature it critiques, Table 4's inversion is genuinely elegant, and the agentic-snooping section speaks directly to ICAIF's moment. As submitted, though, it hands a careful reviewer two internal contradictions on its central claim, an unexplained failed placebo, and

an "a priori" that isn't — at a venue with no rebuttal, any one of those can be fatal. All are fixable within the five days remaining, and the M1 fix in particular leaves the thesis sharper than before: drawdown forecasting rewards *magnitude wherever it lives* — in an asset's own tail or in its co-crash with the market — and nothing else.

Submit after the fixes. Do not submit as-is.